A **July 27, 2026** ANDY HE

The Magical Whip

This is a *variable-mass* system: the moving part of the whip shrinks as it piles up, so Newton's law in the form $\sum F = ma$ does not directly apply. Instead we use the momentum form

$$\dot{P} = F_{\text{ext}},$$

where P is the total vertical momentum of the whip (the pile contributes none, being at rest) and F_{ext} is the total external force on the whip.

The falling column has length $L-y$ and, because every element is in free fall together, a common downward speed $v = \dot{y}$. Its momentum (downward) is

$$P = \lambda\,(L-y)\,v.$$

Differentiating, with $\dot{v} = g$ (free fall) and $\dot{y} = v$,

$$\dot{P} = \lambda\,(L-y)\,\dot{v} - \lambda\,\dot{y}\,v = \lambda\,(L-y)\,g - \lambda\,v^2.$$

The external forces on the whole whip are its weight λLg (downward) and the ground's normal force F_N (upward). Taking downward as positive,

$$F_{\text{ext}} = \lambda Lg - F_N.$$

Thus $\dot{P} = F_{\text{ext}}$ gives

$$\lambda Lg - F_N = \lambda\,(L-y)\,g - \lambda\,v^2 \implies F_N = \lambda\,gy + \lambda\,v^2.$$

Finally, the falling column has descended a distance y, so by free-fall kinematics

$$v^2 = 2gy.$$

Substituting,

$$\boxed{F_N = 3\lambda gy = \frac{3Mgy}{L}}.$$

Interpretation. The term $\lambda gy = Mgy/L$ is the weight of the pile already on the ground; the extra $2\lambda gy$ is the momentum flux —the rate at which the falling whip's momentum is destroyed as each element is brought to rest. This is why the ground feels *three times* the weight of the already-landed portion.